Fairness and Bias in Machine Learning

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The Problem: Systemic Bias

- Systemic *bias* with respect to *gender, race and ethnicity*, often *unconscious*, but prevalent in datasets involving *choices* made by people.

- Some examples include datasets related to human choices in *college admissions, hiring, lending*, or *parole* decisions that discriminate against *African-Americans* or *women*.

Diversity and Meritocracy

- *Diversity* is defined as the practice of including people from a range of *different social and ethnic backgrounds*, and of *different genders and sexual orientations*.

- *Meritocracy* is defined as the practice of selecting people, when a choice is made, *based on achievement* as opposed to wealth or membership in a special social class

Alleviating Bias

- Often in practice, these two goals are in *conflict* and we observe an inherent *trade-off* between them.
- Society has found it challenging to *alleviate such bias* and *achieve diversity* in a way that maintains *meritocracy* in these settings.

Status quo

- In recent years due to the *growing volume of applications* in college admissions, hiring, lending, etc., *machine learning models* are being increasingly employed for such *classification problems*.

- It has been shown that *without appropriate intervention* during training or evaluation, classification models trained on such datasets can be *biased against certain groups of individuals.*

Simple Remedies

- Simple remedies, such as *ignoring the protected attributes*, e.g., gender, race, ethnicity, etc., are largely ineffective due to other factors being *correlated* with the protected attributes.

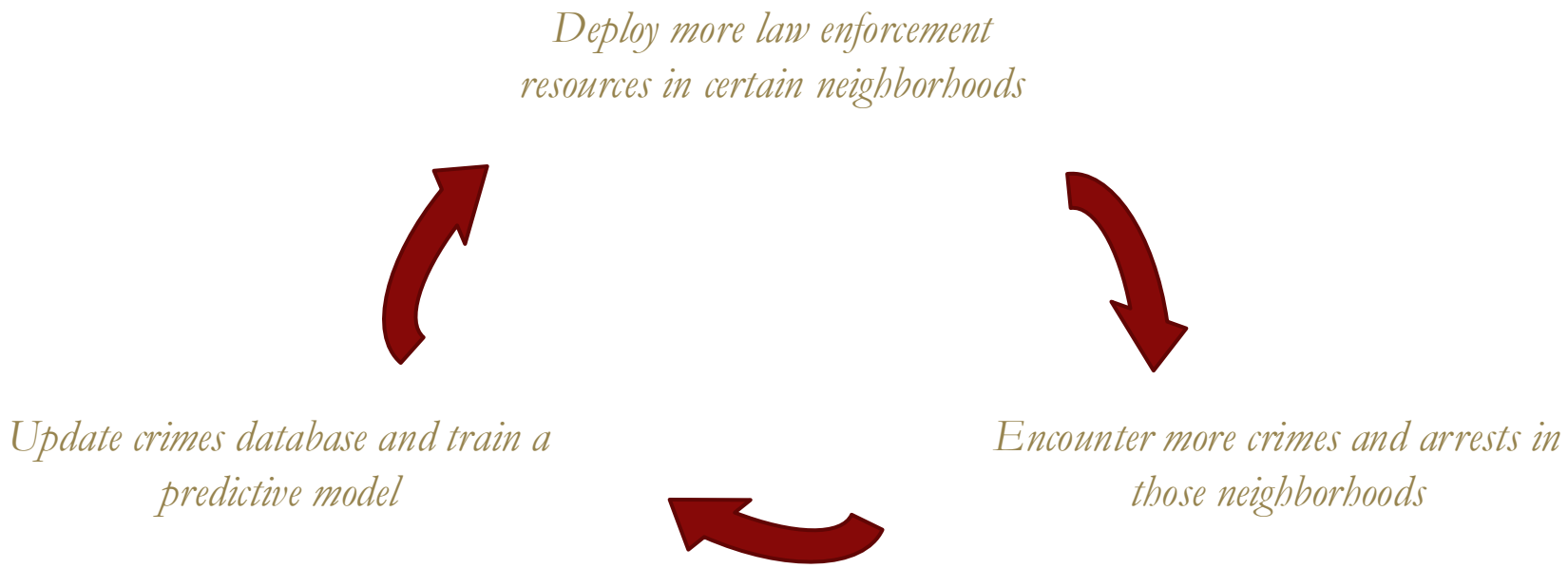
- The data can be inherently biased in possibly complex ways, thus making it difficult to alleviate bias.

The price of inaction

- Classification models that are *actively trained* on such datasets become more *biased over time* through feedback loops wherein the bias against a certain subpopulation can get *amplified*.

- *Feedback loops* have been observed in predictive policing, credit markets and other applications.

Feedback loops (predictive policing)



Law Students Dataset

- A law students dataset that examines *whether the bar exam* taken by law students in the US is *biased against ethnic minorities*, especially black students.

- The dataset also contains anonymized historical information including *age, LSAT scores, law school GPA*, and *undergraduate GPA* .

Notation

- Each of the students is assigned an *outcome* $Y = \{-1, +1\}$ representing either *passing* $(+1)$ or *failing* (-1) the bar exam by the bar council.
  - W: set of white students.
  - B: set of black students.
  - n_w: total number of white students.
  - n_b: total number of black students.
  - p_w: total number of white students who passed the bar exam.
  - p_b: total number of white students who passed the bar exam.

α-biased dataset

- We call a dataset α-biased if the *difference between the rates of positive observations* among a pair of subgroups *W* and *B* based on a protected variable (in this case, race) is at least α.

Definition 1 (α-biased dataset) *A dataset* $\mathcal{X} = \{(x_i, y_i) |\ y_i \in \{-1, 1\}\}$ *is said to be* α*-biased with respect to a pair of subgroups* $\mathcal{W}, \mathcal{B} \subseteq \mathcal{X}$ *if*

$$\left| \frac{\sum_{i \in \mathcal{W}} \mathbb{I}(y_i = +1)}{n_w} - \frac{\sum_{i \in \mathcal{B}} \mathbb{I}(y_i = +1)}{n_b} \right| \geq \alpha.$$

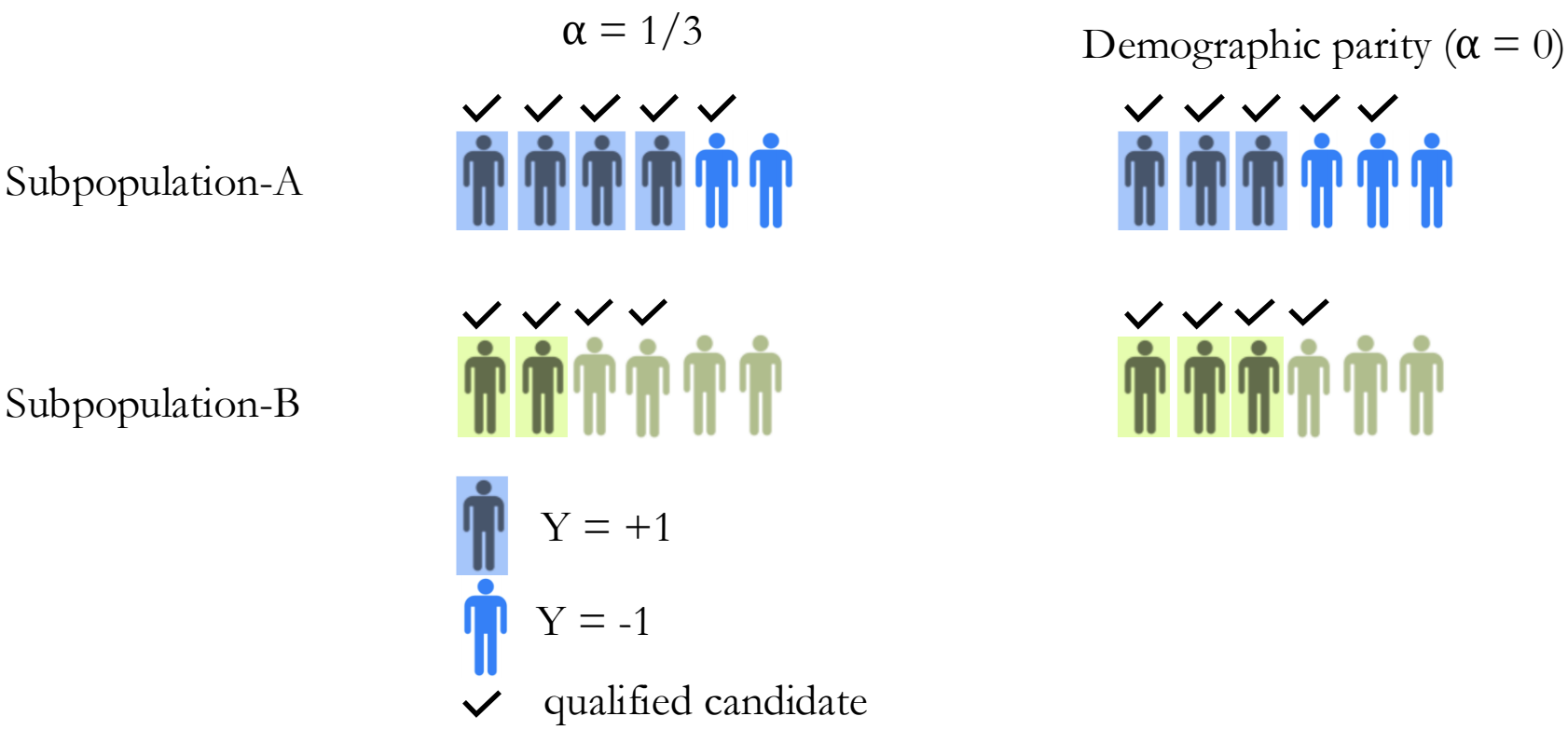
Demographic parity

- Demographic parity imposes the condition that a classifier H should *predict a positive outcome* for individuals across groups with *almost equal frequency*.

Definition 2 (Demographic parity) *A classifier* $\mathcal{H} : X \to \{-1, 1\}$ *achieves demographic parity with bias* ϵ *with respect to groups* $\mathcal{W}, \mathcal{B} \subseteq X$ *if and only if*

$$\left| \frac{\sum_{i \in \mathcal{W}} \mathbb{I}(\mathcal{H}(x_i) = +1)}{n_w} - \frac{\sum_{i \in \mathcal{B}} \mathbb{I}(\mathcal{H}(x_i) = +1)}{n_b} \right| \leq \epsilon.$$

Example of Demographic parity



Proposed solution

- *Flip outcome labels* (Y) while training your model to achieve demographic parity.

- We propose a Mixed-integer Optimization (MIO) problem that achieves this by introducing *binary variables* $z_i \in \{0, 1\}$, $i \in [n]$ to decide which outcome labels to flip.

Proposed solution

- If we decide to *flip the outcome label* of the ith observation: $y_i \in \{-1, 1\}$, the resulting outcome label would be $\tilde{y}_i = y_i(1 - 2z_i)$.

- We define a set of n binary variables (z) that flip at most τ_w proportion of labels in W and τ_b proportion of labels in B given by,

$$\mathcal{Z}_{\tau_w,\tau_b} = \left\{ \mathbf{z} \in \{0,1\}^n : \frac{\sum_{i \in \mathcal{W}} z_i}{n_w} = \tau_w, \ \frac{\sum_{i \in \mathcal{B}} z_i}{n_b} = \tau_b \right\}.$$

Proposed solution

- The parameters τ_w and τ_b are estimated from the data so that the resulting classifier ensures *ε-demographic parity*.

$$\tau_w \leq \frac{n_b \cdot p_w}{n_w(n_w + n_b)} - \frac{p_b}{n_w + n_b} + \frac{n_b \cdot \epsilon}{n_w(n_w + n_b)},$$

$$\tau_b \leq \frac{p_w}{n_w + n_b} - \frac{n_w \cdot p_b}{n_b(n_w + n_b)} - \frac{n_w \cdot \epsilon}{n_b(n_w + n_b)}.$$